

LungCT AI | CLINICAL NOISE ASSESSMENT REPORT

Date: 2026-09-02 21:07:00
System: v1.0.0 (U-Net++ model)

Case / Scan ID:	LCT-74733	Patient Name:	ANONYMOUS (Clinical Study)
Original File Name:	demo_phantom.png	Analysis Protocol:	U-Net++ Noise Segmentation
Scan Modality:	Computed Tomography (CT)	Report Status:	COMPLETED (AI-Generated)

CT Scan Imagery Analysis

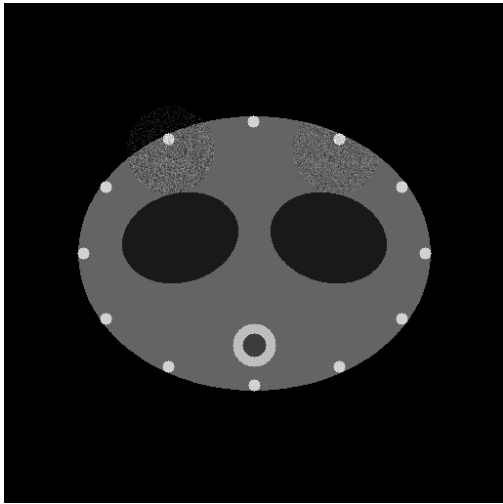


Figure 1: Original CT Scan
Input diagnostic image

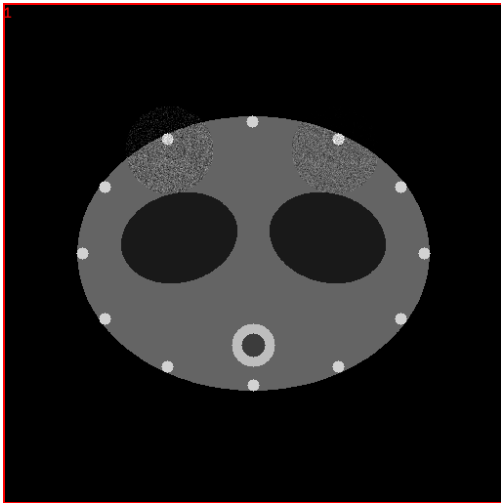


Figure 2: Annotated CT Scan
Red: Gaussian Noise | Blue: Poisson Noise

Quantitative Noise Classification & Severity Metrics

Noise Classification	Pixel Count	Area Coverage (%)	Severity Classification
Gaussian Noise (Electronic)	260,815	99.49%	Critical
Poisson Noise (Photon Shot)	3	0.00%	None
Clean Regions (Signal)	1,326	0.51%	Normal / Target
TOTAL NOISE (Artifacts)	260,818	99.49%	Critical

Clinical Interpretation & Recommendations

Model Findings: CRITICAL noise levels are present in the CT scan, affecting 99.49% of the pixel space. Gaussian noise is 99.49% (Critical) and Poisson noise is 0.00% (None). The signal is severely compromised, rendering the scan clinically unreliable for accurate diagnostic interpretation.

Denoising Recommendations: This scan should be flagged as sub-diagnostic. Recommend immediate patient rescan with adjusted dosage/calibration parameters, or review of the imaging hardware for systemic sensor noise.

Radiologist Review Block	AI System Validation Specs
Reviewing Radiologist Signature	Model Architecture: U-Net++ (Nested UNet) Dice Score: 0.9886 IoU: 0.9778 Precision: 0.9899 Recall: 0.9875

Disclaimer: This report is automatically generated by an artificial intelligence model (U-Net++) for multi-noise classification and severity assessment. It is developed as a Software Engineering Capstone Project. The findings should be reviewed and verified by a licensed radiologist or healthcare professional before clinical decisions are made. All results are for research and educational validation purposes only.